Fiscal Year:	FY 2020	**Task Last Updated:**	FY 03/10/2020
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	GeneLab: Revealing Spaceflight- and Gravity-Response Networks in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
Start Date:	12/19/2016	**End Date:**	12/18/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	25	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/18/2019 per NSSC information (Ed., 9/24/19)		
Key Personnel Changes/Previous PI:	March 2020 report: None		
COI Name (Institution):	Barker, Richard Ph.D. (University of Wisconsin)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	The GeneLab data repository is a powerful resource with which to dissect plant responses to spaceflight at the molecular and genetic level. The research in this GeneLab Innovation study has therefore sought to continue and extend our current analysis of the GeneLab plant science data through bioinformatics, experimental tests of predictions, and development of approaches to facilitate data exploration and rapid sharing of analytical tools. This work directly addresses Goal 1 of the Space Biology program by creating 'new knowledge of how biological systems adapt to space.' In addition, this research approaches some of GeneLab's guiding questions as to how implementation of large scale dataset analyses can help produce new insights and hypotheses about the effects of spaceflight on biological systems. Aim (1): We have been using informatics to ask how spaceflight alters the plant transcriptome and proteome using GeneLab data. We have then compared these changes to responses to other stresses and environments, (e.g., pathogen attack or oxidative stress) where large-scale, publicly available datasets have already been deposited outside of GeneLab. These datasets include responses to partial gravity and gravitropism-related transcriptomes. For example, our analysis of Arabidopsis data from the BRIC (Biological Research in Canisters), EMCS (European Modular Cultivation System), and APEX (Advanced Plant Experiment) experiments has revealed oxidative stress and hypoxia as major flight-related stresses and has targeted gene families ranging from heat-shock proteins to cell wall peroxidases as likely key players in this spaceflight response network. The more comprehensive analysis possible with the bioinformatic tools generated in this innovation study have greatly expanded the range of molecular mechanisms we can detect. Aim (2): The second element of the work has been to take the genes defined from the bioinformatics in Aim 1 and test their function by using mutants in these genes and assessing their responses to a suite of phenotypic assays including: morphometric measurements of gravtiropic and clinostat growth in short and long-term hypoxia and flooding and mechanical stimulation. Inferences from these growth analyses have been compared to those revealed by quantitative PCR results of marker genes identified from the core spaceflight, oxidative stress, hypoxia, and gravitropism transcriptomic analyses. Aim (3): GeneLab datasets also offer the opportunity to address the key challenge of facilitating exploration of these large, complex, and specialized datasets by non-specialists. We have developed a database/exploration environment, the TOAST (Test Of All Space Transcriptomes) database, aimed at increasing such accessibility for the non-specialist. This system uses the Qlik database management software and offers a highly interactive data visualization toolset that allows non-expert users to search across, and explore interconnections within, large and complex datasets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is leading to benefits outside of the spaceflight arena at two levels: (1) tool development and (2) extension of basic knowledge of plant stress responses. At the level of tool development, we have generated a large dataset exploration environment built around the Qlik data visualization toolset. This tool allows rapid comparison and exploration between large-scale 'omics-level datasets, facilitating comparison between transcriptomics/gene expression data and then cross-comparisons to protein-level data such as proteomics analyses. Although focused on spaceflight data within GeneLab's data repository, this architecture is applicable to wider experimental datasets. In parallel we have also been developing gas regulation chambers to allow analysis of plant growth and molecular phenotypes related to lowered oxygen availability. Although again focused on spaceflight-related events of lowered oxygen levels, these devices are also applicable to studying the effects of ground-based phenomena such as flooding effects on plant biology. Our analyses of phenotypes has also extended basic understanding on plant responses to touch, low oxygen levels, and oxidative stress signaling on Earth. For example, we have been able to define an array of genes that when mutated alter responses to oxidative stress on Earth. Oxidative stress is a major element of stress signaling and resistance pathways and so this basic understanding is expanding our broad knowledge of how plants are able to respond to less than optimal environmental conditions.
	Bioinformatics tool development: We have developed, and now deployed, a database and exploration environment for mining the publicly available spaceflight plant biology data called the Test Of Arabidopsis Space Transcriptomes, or TOAST, database. This system is built around the Qlik data visualization software. Thus, we have imported all the GeneLab plant biology data alongside publicly available datasets from US, European, and Chinese researchers into the TOAST database. We have also imported publicly available datasets related to spaceflight-associated stresses including: oxidative stress, responses to growth on the clinostat or during magnetic levitation, and radiation exposure, to help make comparisons to the spaceflight datasets. We have applied a minimal requirement of 3 independent biological replicates for each study allowing for the calculation of statistical significance of any changes seen (P- and Q-values). This has meant the database covers a wide range of studies on Arabidopsis thaliana and Oryza sativa. The database was tested in a public beta-test over 2018-2019 leading to improvements in the user interface, increases in the stability and improvements in the sets of analytical tools available. The database is now in version 4.6 and is publicly available at https:// . Alongside the database an online tutorial has been developed (https://) to aid users in getting started with the system. A manuscript describing the use of the database has also been published in Frontiers in Plant Science. Alongside the TOAST database, a Meta data app has also been deployed to allow a deeper analysis of the available plant spaceflight data. Thus, the meta data associated with each dataset within the TOAST database has been manually curated and imported into an allied Qlik-based database. This approach allows searching of the linked experimental elements such as growth hardware, type of spaceflight, and lighting system used to allow the user to ask which experiments share similar conditions, or to try to understand how the experiments they are analyzing are different. The metadata app is publicly available at: https:// . Mutant analyses: An additional goal for this work has been to use phenotyping of mutants in the genes identified as showing strong links

Task Progress:	to spaceflight-related events from the TOAST-based analysis described above. This phenotyping is being used to ask how well TOAST generates reliable and testable hypotheses about plant responses to spaceflight. Gravitropism, reaction to hypoxia, and response to oxidative stress are all closely linked to spaceflight-related conditions due to spaceflight's disruption in normal plant gravity signaling, reduced availability of oxygen from the lack of convective mixing of gasses in space, and high oxidative loads due to a general background of spaceflight-related stress. These responses have therefore been used to screen the mutants for phenotypes. Analysis using the TOAST database has identified a suite of genes related to oxidative stress such as Heat Shock Proteins (chaperones) and the ROS (reactive oxygen species) producing enzymes such as the respiratory burst oxidase homologs, or RBOHs as likely key players in spaceflight stress responses. Phenotyping mutants in these genes has revealed unexpected linkages to spaceflight-related stresses, such as a role for heat shock proteins in gravitropism and the RBOHs in a broad suite of spaceflight effects from flooding to mechanical responses. TOAST as an educational tool: In addition to a research discovery tool, the TOAST database has also been tested in the educational setting. For example, TOAST has been used in the classroom in Verona and Oregon High Schools in WI, where students went through the process of data analysis, gene discovery, and hypothesis generation. In addition, TOAST has been incorporated into an undergraduate class experience at Carthage College, a 4 year liberal arts college in WI. In collaboration with Dr. Andrea Henle, TOAST analyses were incorporated into her Astrobiology undergraduate class with the goal of leading the students through an authentic research experience, discovering new genes related to spaceflight response in Arabidopsis and then testing these hypotheses with gravitropism assays on mutants in the selected genes. At UWisconsin-Madison, the Principal Investigator (PI) also maintains a "Collaboratory" where biology and engineering undergraduates and high school students come together to develop plant phenotyping equipment and other hardware relevant to our spaceflight-related work. There are generally 10-15 students working on various plant molecular and engineering projects related to space biology and GeneLab with a practical lab course called AstroBotanical Engineering being offered ~each semester. These students have been helping develop the TOAST database interface, using TOAST to analyze spaceflight and related plant datasets, and working on the phenotyping pipelines to test the predictions made through using the GeneLab analyses from the TOAST database.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Barker R, Lombardino J, Rasmussen K, Gilroy S. "Test of Arabidopsis Space Transcriptome: A discovery environment to explore multiple plant biology spaceflight experiments." Front Plant Sci. 2020 Mar 4;11:147. https://doi.org/10.3389/fpls.2020.00147 ; PMID: 32265943; PMCID: PMC7076552 , Mar-2020
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